

DAT42-4

Achieving a ML Proof of Concept

Scoping, building, documenting, and presenting an end-to-end ML project

A proof of concept is a defensible project, not just a working model.

Albert School — 22 hours

Contents

1 Project scoping	2
2 Full exploratory data analysis	2
3 Model building and comparison	2
4 Reproducible pipeline	3
5 Model card	3
6 Production readiness	3
7 Oral presentation	4
8 Peer review	4

1 Project scoping

Scoping turns an open business situation into a project that can be built and judged. It produces a **project brief**. The brief defines:

- **the business problem**: the decision or outcome the project is meant to improve, stated in business terms;
- **the target variable**: the specific quantity the model will predict, and how it is measured and labelled;
- **data availability**: which data exists, where it lives, whether it can be accessed, and whether it is sufficient to predict the target.

Definition 1 A **project brief** is a written document stating the business problem, the target variable, the available data, and what the project will and will not deliver.

The target variable is the bridge between the business problem and the data: it must be both meaningful to the business and recoverable from the data. Assessing data availability before building avoids committing to a target that the data cannot support.

2 Full exploratory data analysis

The project requires a **full EDA** on the project dataset (the systematic univariate, bivariate, and multivariate analysis, data-quality assessment, and visualization established earlier in the programme). Here the EDA has one specific output: it must **document hypotheses that inform the modeling approach**.

Definition 2 A **modeling hypothesis** is a documented claim, grounded in the EDA, about a relationship in the data that should shape how the model is built — for example, which features are likely predictive, which transformations are needed, or which data-quality issues must be handled.

At least **five** such hypotheses must be documented, each tied to evidence from the EDA and each stating how it informs the modeling approach.

3 Model building and comparison

The project must build and evaluate **at least two competing model architectures** on the same problem, then justify the final selection in writing.

Definition 3 A **model comparison** evaluates two or more architectures under the same data, the same evaluation protocol, and the same metric, so that the difference in results is attributable to the models and not to the setup.

The **justified selection** is a written rationale explaining why the chosen architecture is preferred, referring to the comparison results and to any considerations beyond the headline metric (for example interpretability, training cost, or robustness).

4 Reproducible pipeline

The project must deliver a **reproducible pipeline** covering the full path from **raw data ingestion to model output**, and it must be **re-runnable by a different person on a different machine**.

Definition 4 A **reproducible pipeline** is one that another person, on another machine, can run from the raw data to the model output and obtain the same result, without undocumented manual steps.

Reproducibility here is the operational test of the project: the standard is not that the pipeline runs on the author's machine, but that it runs for someone else, from the raw inputs, following only the written instructions.

5 Model card

The project must produce a **model card** documenting the model that was selected.

Definition 5 A **model card** is a structured document describing a model's **intended use**, its **training data**, its **evaluation results**, and its **limitations and risks**.

Each section is required:

- **Intended use**: what the model is for, and the conditions under which its predictions are valid;
- **Training data**: what data the model was trained on, including its scope and known biases;
- **Evaluation results**: how the model performs, under the evaluation protocol used in the comparison;
- **Limitations and risks**: where the model fails or should not be used, and the risks of using it.

6 Production readiness

Beyond the proof of concept, the project must identify **the three most important next steps** to move from PoC toward a production system, and **explain what would change at each step**.

Definition 6 A **production readiness step** names one concrete change required to move from a proof of concept toward a production system, and explains what that change affects (for example data, infrastructure, monitoring, or process).

The deliverable is exactly three such steps, prioritised, each with an explanation of what it changes. The point is to distinguish a proof of concept — which demonstrates feasibility — from a production system, which must be maintained, monitored, and trusted over time.

7 Oral presentation

The project must be **presented in 10 minutes to a mixed technical and non-technical audience**, and the presenter must **answer questions from both sides**.

The constraint is the mixed audience: the presentation must be intelligible to non-technical listeners while remaining defensible to technical ones. Fielding questions from both sides tests whether the project is understood deeply enough to be explained at two levels.

8 Peer review

Each student must conduct a **written peer review of another team's project using a structured evaluation rubric**.

Definition 7 A **structured evaluation rubric** is a fixed set of criteria against which a project is assessed, so that every project is judged on the same dimensions.

The peer review is written and follows the rubric: it assesses the other team's project — its scoping, EDA, model comparison, reproducibility, model card, and production-readiness reasoning — against the shared criteria rather than as a free impression.

Exercises

1. Write a project brief for a real ML project: state the business problem, define the target variable and how it is measured, and assess whether the available data is sufficient to predict it.
2. Conduct a full EDA on the project dataset and document at least five hypotheses, each grounded in EDA evidence and each stating how it informs the modeling approach.
3. Build at least two competing model architectures on the same problem under the same evaluation protocol, and write the rationale for the final selection.
4. Construct the pipeline from raw data ingestion to model output and have a different person run it on a different machine from the written instructions alone.
5. Produce a model card covering intended use, training data, evaluation results, and limitations and risks.
6. Identify the three most important next steps to move the PoC toward production and explain what each step would change.
7. Present the project in 10 minutes to a mixed technical and non-technical audience and answer questions from both sides.
8. Conduct a written peer review of another team's project using the structured rubric.